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Кафедра ИИТ

Лабораторная работа №8
По дисциплине: «Аппаратно-программное обеспечения ЭВМ и сетей»

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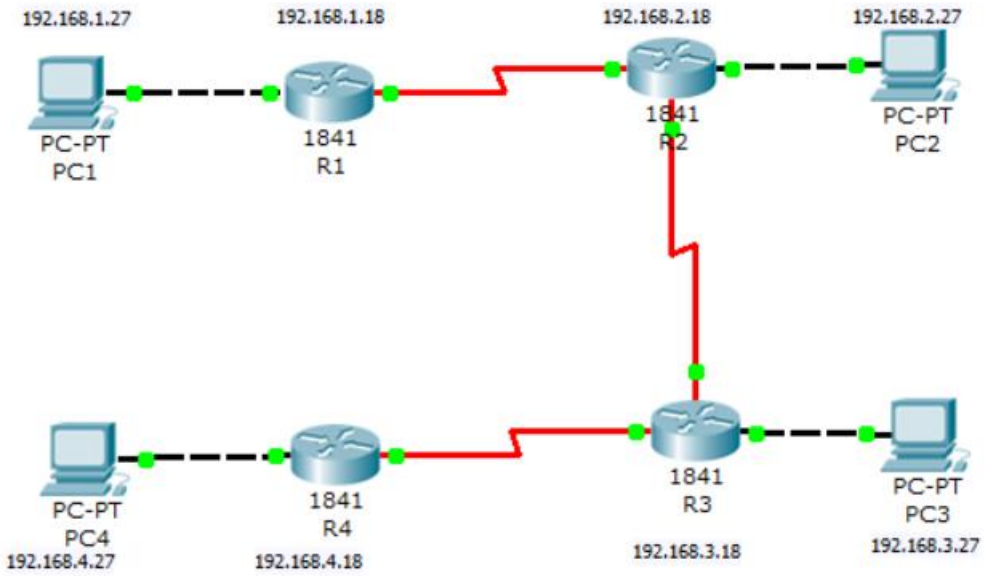
Брест 2026

Цель работы: Настроить динамическую маршрутизацию с помощью протокола RIP на устройствах R1, R2, R3; обеспечить возможность взаимодействия конечных устройств PC1, PC2, PC3 между собой.

Вариант 17

Ход работы

Device	Interface	IP Address	Mask	Default Gateway
R1	Fa0/0	192.168.1.18	255.255.255.0	N/A
	Se0/0/0	10.0.0.1	255.255.255.252	N/A
R2	Fa0/0	192.168.2.18	255.255.255.0	N/A
	Se0/0/0	10.0.0.2	255.255.255.252	N/A
	Se0/0/1	10.0.0.5	255.255.255.252	N/A
R3	Fa0/0	192.168.3.18	255.255.255.0	N/A
	Se0/0/0	10.0.0.6	255.255.255.252	N/A
	Se0/0/1	10.0.0.9	255.255.255.252	N/A
R4	Fa0/0	192.168.4.18	255.255.255.0	N/A
	Se0/0/0	10.0.0.10	255.255.255.252	N/A
PC1	N/A	192.168.1.27	255.255.255.0	192.168.1.18
PC2	N/A	192.168.2.27	255.255.255.0	192.168.2.18
PC3	N/A	192.168.3.27	255.255.255.0	192.168.3.18
PC4	N/A	192.168.4.27	255.255.255.0	192.168.4.18



R1

Physical | Config | CLI

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#no ip domain-lookup
R1(config)#enable secret cisco
R1(config)#line console 0
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#exit
R1(config)#interface fastethernet 0/0
R1(config-if)#ip address 192.168.1.18 255.255.255.0
R1(config-if)#no shutdown

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, change
o up

R1(config-if)#interface serial 0/0/0
R1(config-if)#ip address 10.0.0.1 255.255.255.252
R1(config-if)#clock rate 64000
R1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#
```

R2

Physical | Config | CLI

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#no ip domain-lookup
R2(config)#enable secret cisco
R2(config)#line console 0
R2(config-line)#password cisco
R2(config-line)#login
R2(config-line)#exit
R2(config)#interface fastethernet 0/0
R2(config-if)#ip address 192.168.2.18 255.255.255.0
R2(config-if)#no shutdown

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state t
o up

R2(config-if)#interface serial 0/0/0
R2(config-if)#ip address 10.0.0.2 255.255.255.252
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up


R2(config-if)#interface serial 0/0/1
R2(config-if)#ip address 10.0.0.5 255.255.255.252
R2(config-if)#clock rate 64000
R2(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
R2(config-if)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R2#
```

R3

Physical | Config | CLI

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R3
R3(config)#no ip domain-lookup
R3(config)#enable secret cisco
R3(config)#line console 0
R3(config-line)#password cisco
R3(config-line)#login
R3(config-line)#exit
R3(config)#interface fastethernet 0/0
R3(config-if)#ip address 192.168.3.18 255.255.255.0
R3(config-if)#no shutdown

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state t
o up

R3(config-if)#interface serial 0/0/0
R3(config-if)#ip address 10.0.0.6 255.255.255.252
R3(config-if)#no shutdown

R3(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up


R3(config-if)#interface serial 0/0/1
R3(config-if)#ip address 10.0.0.9 255.255.255.252
R3(config-if)#clock rate 64000
R3(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
R3(config-if)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R3#
```

R4

Physical | Config | CLI

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R4
R4(config)#no ip domain-lookup
R4(config)#enable secret cisco
R4(config)#line console 0
R4(config-line)#password cisco
R4(config-line)#login
R4(config-line)#exit
R4(config)#interface fastethernet 0/0
R4(config-if)#ip address 192.168.4.18 255.255.255.0
R4(config-if)#no shutdown

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed
o up

R4(config-if)#interface serial 0/0/0
R4(config-if)#ip address 10.0.0.10 255.255.255.252
R4(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
R4(config-if)#end
R4#
%SYS-5-CONFIG_I: Configured from console by console

R4#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R4#
```

R1

Physical Config CLI

IOS Command Line Interface

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol


FastEthernet0/0        192.168.1.18    YES manual up                    up

FastEthernet0/1        unassigned      YES unset  administratively down down

Serial0/0/0            10.0.0.1        YES manual up                    up

Serial0/0/1            unassigned      YES unset  administratively down down

Vlan1                  unassigned      YES unset  administratively down down
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 1 subnets
C       10.0.0.0 is directly connected, Serial0/0/0
C    192.168.1.0/24 is directly connected, FastEthernet0/0
R1#
```

R2

Physical Config CLI

IOS Command Line Interface

```
R2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protoco


FastEthernet0/0        192.168.2.18    YES manual up                    up

FastEthernet0/1        unassigned      YES unset  administratively down down

Serial0/0/0            10.0.0.2        YES manual up                    up

Serial0/0/1            10.0.0.5        YES manual up                    up

Vlan1                  unassigned      YES unset  administratively down down
R2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 2 subnets
C       10.0.0.0 is directly connected, Serial0/0/0
C       10.0.0.4 is directly connected, Serial0/0/1
C    192.168.2.0/24 is directly connected, FastEthernet0/0
R2#
```

R3

Physical Config CLI

IOS Command Line Interface

```
R3#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocc


FastEthernet0/0        192.168.3.18    YES manual up                    up

FastEthernet0/1        unassigned      YES unset  administratively down down

Serial0/0/0            10.0.0.6        YES manual up                    up

Serial0/0/1            10.0.0.9        YES manual up                    up

Vlan1                  unassigned      YES unset  administratively down down
R3#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 2 subnets
C       10.0.0.4 is directly connected, Serial0/0/0
C       10.0.0.8 is directly connected, Serial0/0/1
C    192.168.3.0/24 is directly connected, FastEthernet0/0
R3#
```

R4

Physical Config CLI

IOS Command Line Interface

```
R4#show ip interface brief
Interface              IP-Address      OK? Method Status                Protoc


FastEthernet0/0        192.168.4.18    YES manual up                    up

FastEthernet0/1        unassigned      YES unset  administratively down down

Serial0/0/0            unassigned      YES manual administratively down down

Serial0/0/1            10.0.0.10       YES manual up                    up

Vlan1                  unassigned      YES unset  administratively down down
R4#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter are
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 1 subnets
C       10.0.0.8 is directly connected, Serial0/0/1
C    192.168.4.0/24 is directly connected, FastEthernet0/0
R4#
```

R1

Physical Config CLI

IOS Comma

```
R1(config)#router rip
R1(config-router)#network 192.168.1.0
R1(config-router)#network 10.0.0.0
R1(config-router)#end
```

R2

Physical Config CLI

IOS Comma

```
R2(config)#router rip
R2(config-router)#network 192.168.2.0
R2(config-router)#network 10.0.0.0
R2(config-router)#end
```

R3

Physical Config CLI

IOS Comm

```
R3(config)#router rip
R3(config-router)#network 192.168.3.0
R3(config-router)#network 10.0.0.0
R3(config-router)#end
```

R4

Physical Config CLI

IOS Command

```
R4(config)#router rip
R4(config-router)#network 192.168.4.0
R4(config-router)#network 10.0.0.0
R4(config-router)#end
```

R1

Physical Config CLI

IOS Command Line Interface

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 3 subnets
C       10.0.0.0 is directly connected, Serial0/0/0
R       10.0.0.4 [120/1] via 10.0.0.2, 00:00:15, Serial0/0/0
R       10.0.0.8 [120/2] via 10.0.0.2, 00:00:15, Serial0/0/0
C    192.168.1.0/24 is directly connected, FastEthernet0/0
R    192.168.2.0/24 [120/1] via 10.0.0.2, 00:00:15, Serial0/0/0
R    192.168.3.0/24 [120/2] via 10.0.0.2, 00:00:15, Serial0/0/0
R    192.168.4.0/24 [120/3] via 10.0.0.2, 00:00:15, Serial0/0/0
```

R2

Physical Config CLI

IOS Command Line Interface

```
R2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 3 subnets
C       10.0.0.0 is directly connected, Serial0/0/0
C       10.0.0.4 is directly connected, Serial0/0/1
R       10.0.0.8 [120/1] via 10.0.0.6, 00:00:06, Serial0/0/1
R    192.168.1.0/24 [120/1] via 10.0.0.1, 00:00:11, Serial0/0/0
C    192.168.2.0/24 is directly connected, FastEthernet0/0
R    192.168.3.0/24 [120/1] via 10.0.0.6, 00:00:06, Serial0/0/1
R    192.168.4.0/24 [120/2] via 10.0.0.6, 00:00:06, Serial0/0/1
```

R3

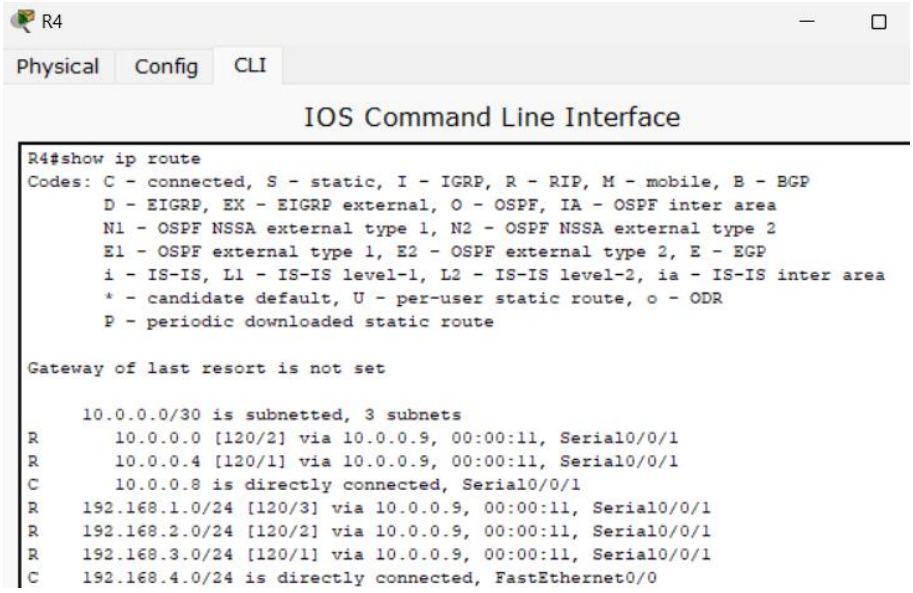
Physical Config CLI

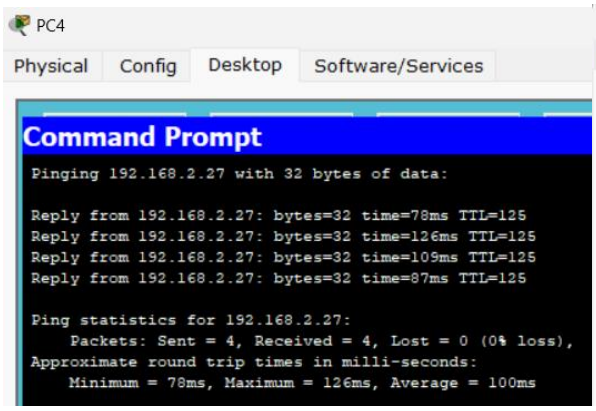
IOS Command Line Interface

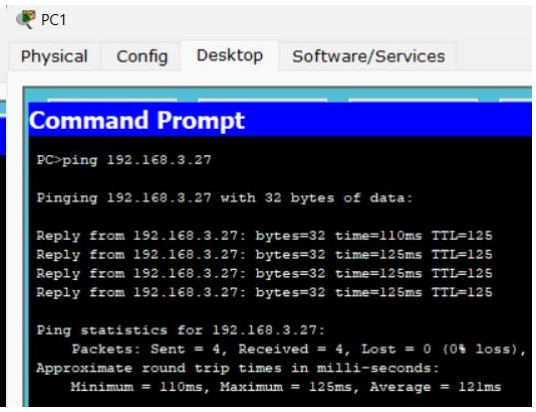
```
R3#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

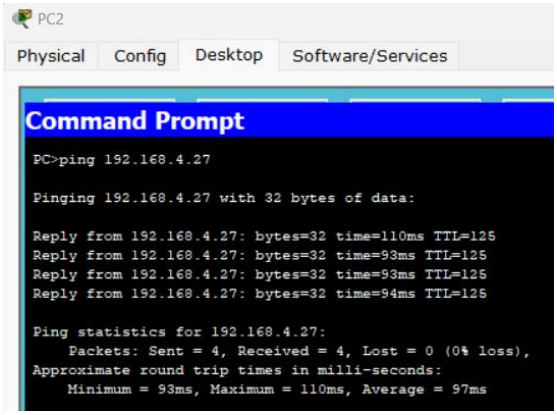
Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 3 subnets
R       10.0.0.0 [120/1] via 10.0.0.5, 00:00:07, Serial0/0/0
C       10.0.0.4 is directly connected, Serial0/0/0
C       10.0.0.8 is directly connected, Serial0/0/1
R    192.168.1.0/24 [120/2] via 10.0.0.5, 00:00:07, Serial0/0/0
R    192.168.2.0/24 [120/1] via 10.0.0.5, 00:00:07, Serial0/0/0
C    192.168.3.0/24 is directly connected, FastEthernet0/0
R    192.168.4.0/24 [120/1] via 10.0.0.10, 00:00:26, Serial0/0/1
```









Вывод: я настроила динамическую маршрутизацию с помощью протокола RIP на устройствах R1, R2, R3; обеспечила возможность взаимодействия конечных устройств PC1, PC2, PC3 между собой.